

Seventh AI Vision

The complete feature inventory — AI video surveillance, security intelligence and guard workforce management in one multi-tenant platform, built for Singapore security agencies.

274 DATABASE TABLES	997 API OPERATIONS	102 API MODULES	106 WEB ROUTES	52 MOBILE SCREENS	11 AI DETECTION MODULES
197 PERMISSIONS	8 BUILT-IN ROLES	151 MIGRATIONS	5,920 AUTOMATED TESTS	4 LANGUAGES	3 CLIENT APPS

What this document is

Every capability listed here was read from the source — the API routers, the database migrations, the services and the client pages — not from a specification of what was intended. Where a feature has a documented limitation, this document says so rather than omitting it. What is still being built is listed separately, under *In progress and planned*, and is not counted in the figures above.

The platform is one system whose parts share a tenant, a permission model and an audit trail: AI-driven video surveillance, the security intelligence that relates what the cameras, drones, patrols and guards report, and the workforce management that turns an alert into someone actually attending. Each part runs without the others — a guarding-only customer needs no cameras, and a camera-only customer needs no roster — but the value is in the join.

One rule runs through all of it: AI detects, relates, assesses and recommends. An authorised person makes the security decision.

About this edition

This is a living document. It is rebuilt from a script kept beside it in the repository (`scripts/make_feature_pdf.py`) each time the platform moves, so it is regenerated rather than hand-edited. A feature appears in the numbered modules only once it is merged, tested and running; until then it is in the table of what is in progress and planned.

Edition	Date	Platform at	What it covers
Edition 1	9 September 2026	0100	The first reference: thirteen modules, from identity to compliance.
Edition 2	7 October 2026	0148	Adds the vendor's platform console, Virtual Patrolling, Drone Patrol, AI Security Intelligence, and the first six phases of the enterprise expansion: investigation, evidence packages, the security map, guard response with SLA and escalation, the occurrence book reviewed and summarised, and the SOP library. Adds the sections on where AI is used, what is in progress, and known limits.
Edition 3	8 October 2026	0150	Adds phases 7 and 8 of the enterprise expansion: visitor and contractor authorisation (module 23), and device health, the asset register and maintenance work orders (module 24). The vendor's console becomes module 25.
Edition 4	8 October 2026	0151	Adds phase 9 of the enterprise expansion: risk patterns and advice (module 25) — where what went wrong gathers, what stands out with how much history it rests on, and a person's answer. The vendor's console becomes module 26.

How the document is laid out

- **At a glance** — one page for the person who signs.
- **Part 1, modules 01 to 13** — the platform as the first edition described it, brought up to date.
- **Part 2, modules 14 to 26** — what has been added since: patrols from a screen and from the air, security intelligence, and the enterprise expansion.
- **Where AI is used, and where a person decides** — every place a model or a rule does work, what it is trusted with, and what stays a person's. **AI capabilities added or upgraded later are recorded here.**
- **In progress and planned** — what is being built now and what follows.
- **Client applications, roles, Singapore, and known limits.**

At a glance

A security agency runs on four or five systems that do not talk to each other: a spreadsheet roster, a messaging group for incidents, a paper occurrence book at every guardhouse, a payroll spreadsheet, and a filing cabinet for whatever a client asks about later. This is one system that replaces all of them.

What changes

	Today	With Seventh AI Vision
Client asks for proof of a patrol	Phone calls, a photo of a page	On screen in seconds — or the client checks their own login
A key is unaccounted for	Nobody knows until it is needed	Shown at every handover, with who took it and when
Leave approved on a live roster	Someone remembers, or nobody does	The uncovered post queues until it is filled
Month-end payroll	Hours retyped from timesheets	From real check-ins, supervisor-approved first
A lone officer collapses	Nothing, until somebody notices	Man-down alerts the control room with their last position
Invoicing a client	Estimated from the contract	From the hours actually worked at their site
Five alerts from one event	Five things to read, on five screens	One situation, with what is unusual about it and a suggested response
"Where was that vehicle seen?"	Somebody scrubs through recordings	Every sighting of the plate, in order, with time and distance between
Evidence asked for months later	Whatever has not been purged	A sealed package under a hold, with its chain of custody
A guard is sent to an incident	A call, and hoping	Accepted, on the way, arrived, reported — each recorded, and late ones escalated

The platform in twenty-five lines

- **Identity and access** — 8 roles, 197 permissions, SSO, 2FA, audit log, per-tenant branding, four languages.
- **Video surveillance** — cameras, live wall, recording with checksums, playback, privacy zones, NVR.
- **AI analytics** — eleven detection modules, licensed individually.
- **Alerts and incidents** — triage, dispatch, escalation, emergency broadcast.
- **Roster** — a whole month auto-scheduled against nine rules, reviewed as a draft before it goes live.
- **Attendance** — face check-in with liveness, geofenced, eight live states.
- **Guard operations** — patrol scanning, occurrence book, post orders, panic button, man down.
- **Guardhouse registers** — keys, lost & found, defects, equipment, uniform, and a handover two people sign.
- **Visitor management** — pre-registration, passes, contractors, deliveries.
- **Physical security and IoT** — access control, alarms, barriers, parking.
- **HR and payroll** — leave, training, timesheets, CPF, overtime, public holidays, PWM grades and wage floors, IR8A.
- **Commercial** — invoices from real hours, and a client portal.
- **Compliance** — chain of custody, PDPA, retention, scheduled reports.

- **Virtual Patrolling** — a scheduled camera-by-camera inspection from a screen, with a report.
- **Drone Patrol** — planned flights whose camera the AI watches like any other.
- **AI Security Intelligence** — events related into situations, assessed, with a suggestion a person decides on.
- **Smart Investigation** — one search across fourteen sources, and the trail of a plate or a watchlist face.
- **Evidence packages** — sealed, checksummed, held against deletion, exported.
- **Security map** — cameras, guards, drones, incidents and the places of a site on one map.
- **Guard response** — the steps of a response, three SLA clocks, escalation policies.
- **Occurrence book, reviewed** — search, supervisor review, correction, standing instructions, a shift summary.
- **SOP library** — versioned, approved procedures, quoted beside the incident.
- **Visitor authorisation** — a host's yes or no to a visit, for where and for how long, read at the gate.
- **Assets and maintenance** — device health as devices report it, an asset register, and work orders a person accepts.
- **Platform console** — the vendor's own: customers, pricing, invoicing, errors, support sessions.

Built for Singapore, not adapted to it

PLRD licence expiry per officer, and what a site requires an officer to hold. PWM grades across the seven-grade ladder, with wage floors pay is checked against. CPF by age band against the Ordinary Wage ceiling. Gazetted public holidays and holiday pay. The 72-hour monthly overtime cap flagged on every run. IR8A. PDPA consent and data subject requests. An occurrence book built against the Private Security Industry Act. SGD invoicing. English, Chinese, Malay and Tamil. Runs hosted or on your own servers.

Part 1 — Platform capabilities

The thirteen modules of the first edition, brought up to date. Where a module has grown into one of its own since, it says where to read on.

01. Multi-tenancy, identity and access

One installation serves many security agencies, isolated in the database rather than in application code.

- **Row-level security on every table.** Tenant isolation is a Postgres policy, forced even on the table owner, not a WHERE clause a developer can forget. Records that must not be rewritten are granted insert and read only, and the application's database role cannot empty a table.
- **Eight built-in roles** — Super Admin, Admin, Manager, Supervisor, Operator, Security Guard, Viewer, Client — plus custom roles defined per tenant, across 197 permission codes.
- **Site scope.** A user can be held to particular sites; every list, search and record honours it.
- **The platform operator is not a customer's administrator.** Super Admin runs the platform from a tenant of its own and holds none of a customer's day-to-day permissions (module 26).
- **Single sign-on and SCIM** for automated user provisioning and de-provisioning from the customer's own directory.
- **Two-factor authentication (TOTP)** — required, not optional, for the platform owner — with IP allow-listing, API keys, rate limits, and per-user session management with forced revocation and account unlock.
- **Hash-chained audit log** covering every privileged action, with the actor, role, tenant, site, result and request id, and verifiable end to end.
- **Per-tenant branding** — company name, logo and primary colour — with the product name held separately so white-labelling never hides which platform is running.
- **Per-tenant timezone** applied to every date the application renders, so a Singapore operator never reads a UTC timestamp.
- **Four languages:** English, Chinese, Malay and Tamil.

02. Video surveillance

Cameras, streams and recordings, with the operator surfaces built around watching many at once.

- **Camera management** with ONVIF discovery, PTZ control, and RTSP credential validation before a camera is saved rather than after it fails.
- **Live Wall** — a multi-camera video wall with saved layouts and profiles, and genuine pop-out windows for multi-screen control rooms.
- **Continuous recording** with retention policies that inherit tenant to site to camera, so an exception is set once where it applies.
- **Recording integrity.** Each recording carries a checksum that is verified, and the platform's health check reports how many streams should be recording against how many are.
- **Playback and a recordings library**, with export.
- **Privacy zones** masked out of both live view and recordings.
- **NVR integration** and HLS streaming for browser playback.
- **Site map view** plotting sites with live camera health and manning (the full security map is module 19).
- **Detection heat maps** and trend analytics per camera and site.

03. AI analytics

Eleven detection modules, licensed individually per tenant so a customer pays for what they run.

- **Licence plate recognition (LPR/ANPR)** — entry and exit lanes, watchlist matching, vehicle journeys.
- **Face recognition** with watchlists and pgvector similarity matching.
- **Intrusion detection** against drawn restricted zones.
- **PPE compliance** — helmet, vest and equipment checks.
- **Crowd density** and zone occupancy.
- **Fire and smoke detection.**
- **Weapon detection.**
- **Behaviour analysis** for anomalous movement.
- **Camera tampering** — obstruction, defocus and repositioning.
- **Abandoned object detection.**
- **Fall detection** for slips and collapses in camera view.
- The same detectors watch a drone's camera (module 15), and what they report is related and assessed by the intelligence layer (module 16).

04. Alerts, incidents and response

Turning a detection into a decision, and a decision into an attended response.

- **Alert rules engine** with per-module thresholds, schedules and site scoping.
- **Deduplication rules** so one event does not become forty alerts.
- **Alert routing** to the right people by role, site and severity.
- **Alerts that are not about a camera** — a contractor's permit expiring, a visitor overstaying, a guard pressing panic — carry their own site, so a guarding-only agency with no cameras receives them, and they reach the people responsible for that site.
- **Incident management** — assignment, notes, severity, status and resolution, with bulk actions.
- **Command Centre** — the live operational picture across every site.
- **Action Centre** — cross-role duty guidance telling each person what needs them next.
- **Emergency broadcast** to all staff, a role or a site, with per-recipient acknowledgement tracking.
- **Dispatch** for sending a named guard and tracking arrival. The guard's own steps, the SLA clocks and escalation are module 20.
- **Notification channels and rules** — email, SMS, webhook, in-app and push, with a delivery log.

05. Roster and scheduling

Building the roster, and keeping it true when reality interferes.

- **Shift patterns** per site, guard and weekday, expanded into concrete shifts on a rolling window.
- **Auto-scheduler** scoring candidates against nine rules: minimum rest (11 hours), maximum consecutive days (6), day/night balance, stated preference, approved leave, the site's own duty team, minimum coverage, supervisor presence, and off-day stagger.
- **Per-shift-type duty strength.** A site is staffed separately for day and night duty, and the scheduler fills every post — not one guard and a warning.
- **Duty Teams** — named day and night teams per site, with the shortest-staffed sites sorted first. The scheduler draws from the team, falling back to the wider pool rather than leaving a post empty, and flags when it had to.

- **One post per guard.** An Operator or Security Guard holds one posting; Supervisors and Managers may cover several sites. The database holds the rule, so the same officer cannot be put on three sites at once.
- **What a site requires an officer to hold.** A tenant-wide baseline (every officer holds a valid licence) plus what a particular client's contract adds. An officer who does not meet it is reported daily — it is not a block, because a manager covering a 2 am no-show must not be stopped by the roster tool.
- **Guard shift preferences** — preferred shift type and off days, set on the guard's own record and honoured by the scheduler.
- **Draft rosters** reviewed and edited before publishing, with warnings for unfilled posts, coverage shortfalls and missing supervisors.
- **Schedules a whole calendar month in one action**, or any range up to 62 days, with the rules configured per run rather than read out of the source.
- **Cover requests.** When approved leave lands on a published shift, the uncovered post joins a queue that stays open until a supervisor assigns a replacement or records that none is needed. The system refuses a replacement who is on leave or already rostered on an overlapping shift.

06. Attendance

Proving a guard was where they were rostered, when they were rostered.

- **Selfie check-in and check-out** with liveness scoring to reject a photograph of a photograph.
- **Anti-spoof location** — mock-GPS detection flagged on the record.
- **Site geofence** as a radius or a drawn polygon, with off-site check-ins marked rather than silently accepted.
- **Per-site late grace** — a remote gate with one bus an hour cannot hold the same standard as a lobby on a train line.
- **Live attendance board** with eight states: on time, late, on break, due now, not reported, upcoming shift, approved leave and checked out.
- **Escalation ordering** — sites with guards due and nobody reported sort to the top under a no-guard-on-site banner.
- **Correction queue** for supervisor-approved amendments.

07. Guard operations

What a guard actually does on shift, and the record it leaves.

- **Guard tours** — routes and checkpoints scanned by QR or NFC, with tour schedules and compliance reporting.
- **Digital Occurrence Book** — entries with immutable timestamps, built against the Private Security Industry Act requirement. Fifteen kinds of entry. Search, supervisor review and correction are module 21.
- **Post orders** per site, with per-guard acknowledgement so there is a record of who read what. Versioned, approved procedures are module 22.
- **Violations** raised against a guard, with review.
- **GPS tracking** and geofence entry/exit events.
- **SOS panic button** — writes a critical occurrence entry, raises an incident, and pushes a real-time alert to every supervisor watching.
- **Body-worn cameras** — assignment, recordings and events.
- **Man down.** The guard's phone watches its accelerometer and raises an alert when they stop moving, or are knocked and then lie still. A countdown they can cancel avoids a phone left on a desk waking the control room — but the deadline is held on the server, so a handset that shattered on impact, ran flat or lost signal

still gets help sent. Off by default; the thresholds are a tenant setting.

08. The guardhouse registers

The books kept on paper at almost every site — the ones a client audit asks to see first, and the ones nobody can report on.

- **Key register.** A cabinet per site, and a log of who took which key, when, and whether it came back. Built so the useful question — what is still out right now — is answered directly, and the database refuses to let one key be issued twice. Keys go to contractors and tenants as often as to staff, so a holder can be a named person without an account.
- **Lost and found.** What was handed in, where it is stored, and who took it away. Only the last few characters of a claimant's identity document are ever stored — the check that matters is against the card in their hand. Property held past the retention period is reported, because an item kept indefinitely is a PDPA problem accruing quietly.
- **Facility defect log.** The blown stairwell light, the leak in the car park. The status ladder stops where a security company's control does: the record carries who it was referred to and the building's own ticket number, so chasing it is possible. Safety hazards and anything open past a fortnight are surfaced separately.
- **Equipment register.** Radios, torches, batons and body cameras signed out to a named officer. The condition it comes back in writes through to the item, so the next officer is not issued a broken torch.
- **Uniform register.** Issued by quantity and size, with partial returns — a guard who resigns hands back three of four shirts — and the deposit held against the kit.
- **One question answers both:** what is this officer holding? The screen a supervisor opens on somebody's last day.
- **Structured shift handover.** Two signatures, not one person's note. The outgoing guard completes a checklist configured per site; the incoming guard accepts it or DISPUTES it. A counted item is measured against the register, so eleven keys against a handover claiming twelve is visible rather than signed for. Every handover carries the keys still out, property held, kit signed out and defects open.

09. Visitor management (VMS)

The gatehouse: every visitor, not just the ones in vehicles.

- **All visit types** — walk-in, delivery, drop-off and pick-up — on one board.
- **Per-site form configuration.** Each site defines its own fields, types and required flags.
- **Pre-registration** as a dialog on the same page, so the gatehouse never navigates away mid-queue.
- **Fast LPR entry** — the plate is read and the record opened with known details already filled, because a vehicle at the barrier blocks the road while anyone types.
- **Visitor labels with QR**, 62 mm, printed at the operator's discretion rather than automatically.
- **QR check-out** — scanning the label closes that visit.
- **Exit-camera auto check-out** from an LPR read.
- **Automatic day-close** so a visit never runs past its own day.
- **Full action log** naming the operator behind every entry and exit.
- **Contractors** — vetting, accreditation, work permits with approval and a safety briefing, and deliveries.
- *Being built now:* host approval of a visit, the places it is authorised for, escort, and what the gate is told — see *In progress and planned*.

10. Physical security and IoT

The hardware around the guard.

- **Access control** — doors, credentials, rules and an event log.
- **Barrier control** with commanded open/close and an audit of who commanded it.
- **Alarm panels and zones** with event history.
- **IoT sensors** — readings, thresholds and alerts.
- **Car park management** — car parks, bays, zones, tariffs, sessions and dedicated LPR cameras, with a free-parking allowance per site.
- **Device protocol support** for third-party integration.

11. HR, leave, training and payroll

The workforce behind the roster.

- **Employee records** — NRIC/FIN, date of birth, nationality, work pass type and expiry, address, emergency contact, bank details and designation.
- **Document management** with expiry tracking and expiring-soon warnings on passports, work passes and certifications.
- **Leave** — configurable types, entitlement balances, requests with supporting documents, and approval that feeds the roster directly.
- **Training** — courses, a real quiz engine with question banks, attempts and pass records.
- **Certifications** with issuing body, number and expiry.
- **Payroll** — CPF by age band with the S\$7,400 Ordinary Wage ceiling, work-pass eligibility, overtime at 1.5x, and monthly, daily or hourly pay bases.
- **Timesheet approval.** A human between a check-in scan and a payslip. Hours are snapshotted when submitted, so an approval stays a signature on a fixed document rather than on one that keeps changing underneath. Payroll always reports which guards have no approved timesheet, and can be set to refuse to pay them.
- **Public holidays and shift allowances.** A gazetted holiday calendar per tenant, holiday pay, and named allowances attached to a shift type.
- **Overtime cap.** Hours beyond the 72-hour monthly statutory limit are flagged on the run rather than paid silently.
- **PWM grade and wage floors.** The grade is on the officer record across the seven-grade ladder, and pay is checked against a floor: the statutory one, held once for every customer, or a company's own, which may only be higher. A floor is judged as of the payroll period, so restating March is judged against March. Where no rate has been loaded the result is *not assessed*, never *compliant*.
- **PLRD licence number and expiry** tracked per officer.
- **Payslip PDFs** and an **IR8A** annual summary and PDF.
- *Documented limitations:* graduated first- and second-year PR CPF rates and the Additional Wage ceiling are not implemented; no PWM rates are shipped — they are loaded from the published schedule with its source.

12. Commercial

Getting paid for the hours actually worked.

- **Client management** with contacts and billing addresses.
- **Invoices generated from real attendance** — the loop closes from roster to check-in to worked hours to invoice line, per site.
- **Per-site bill rates** with overtime, tax and invoice PDFs.
- **Client portal** — the customer signs in and sees their own sites, incidents and invoices, and nothing else.
- **What the vendor charges the agency** — plans, module prices, subscriptions and the vendor's own invoices — is module 26.

13. Compliance, evidence and reporting

The record that survives an audit.

- **Evidence management** with an access log recording every view and download. Sealed packages, holds and the full chain of custody are module 18.
- **PDPA** — consent records and data subject access requests, with erasure and subject export.
- **Contractor accreditation** and work permit tracking, with expiry warnings.
- **Data retention policies** per tenant, site and camera, and a hold that stops a purge.
- **Scheduled reports** delivered on a recurring basis by email or webhook.
- **Checksums on what is handed over** — patrol snapshots, patrol reports, recordings and evidence files — so a document can be checked later against what was written.
- **Exports** in operational formats, and a full analytics suite.

Part 2 — Added since the first edition

Thirteen modules built after the first edition. Each was added beside what already worked: new tables, new API modules and new screens, with the existing ones left as they were. Every one keeps the same rule — the platform detects, relates, assesses and recommends, and a person decides.

ADDED SINCE EDITION 1 · MIGRATIONS 0116, 0117, 0121

14. Virtual Patrolling

A scheduled, camera-by-camera inspection carried out from a screen — the patrol a control room does without leaving it.

- **Schedules** name a site, an ordered list of cameras and a set of questions. When its time arrives — in the site's own time zone — a session is created for the assigned officer.
- **A real snapshot at every camera.** The officer works through the cameras, a frame is captured from the live stream at each, and the questions are answered against it. Each snapshot carries a checksum.
- **What was asked is frozen into the session.** Editing a schedule mid-patrol does not change what the officer is being asked, and a patrol from March still reads as it did in March.
- **A failed answer can raise an incident** in the platform's own incident system.
- **One execution, one session** — enforced by the database, so two workers or a restart cannot create it twice. A patrol nobody did is marked missed.
- **A report for every patrol** — PDF and an Excel workbook, stored with a checksum and emailed at once or as a daily, weekly or monthly digest, each period sent once.
- **A Command Centre panel** for patrols due, running and missed.

ADDED SINCE EDITION 1 · MIGRATIONS 0123 TO 0130

15. Drone Patrol

An autonomous drone security patrol built into the platform rather than beside it: the drone's camera is one of the platform's cameras.

- **Fleet, routes, zones and missions.** A drone is assigned to a site and flies a planned route on a schedule or on request, with pre-flight checks, a command queue, and maintenance logs.
- **The same AI watches it.** The detectors that watch fixed cameras watch the drone's. What they see is judged in the context of where the drone was and what that place is, and becomes a drone security event with a risk level.
- **Checked against fixed cameras.** The platform works out which fixed cameras could have seen the same thing, and whether they did.
- **Into the platform's own incident and dispatch.** A serious event becomes an incident, and a guard is sent through the existing dispatch.
- **Verify with drone.** An officer can ask for a flying look at something a fixed camera or a guard reported.
- **Site edge gateway.** A site can keep flying while its link to the platform is down: the gateway buffers what happened and syncs it exactly once when the link returns.
- **A record of every flight** — a PDF and a workbook, emailed on a schedule — with analytics, an analytical risk map and recommendations.
- **Licensed per tenant**, with limits on drones, missions and sites; sixteen permissions of its own; web and phone screens.
- *Documented limitation:* everything here has been proven against a simulator. No physical drone, manufacturer SDK, flight controller or real video stream has been connected.

16. AI Security Intelligence and human decision support

One flow over every source: events are brought to one shape, related into situations, assessed, and a response is suggested — and then a person decides.

- **One shape for every security event** — CCTV, LPR, face, drone, virtual patrol, alarm, sensor, guard and system events — read from the tables that already hold them, without changing any of them.
- **What a site expects.** Site and camera profiles state what is normal there and when, and planned activity (a work permit, an expected visitor) is read as context.
- **Situations.** Events that belong together — the same plate, the same watchlist entry, neighbouring cameras moments apart — become one situation instead of several alerts.
- **How unusual, and how much it matters.** Each situation is assessed for normality and risk, with the reasons kept beside the score.
- **A suggestion, not an action.** The layer recommends a response and says what the recommendation rests on.
- **A person decides.** The suggestion, the human decision and the action carried out are three separate records. Only a decision made by a signed-in person causes anything to happen, and it happens through the platform's existing functions — dispatch, incident, a drone look.
- **Reports from the ground.** A guard on the phone accepts, arrives and reports what they found; a drone's look comes back as an event and the situation is assessed again.
- **A unified timeline and the evidence** of each situation, with an AI-assisted summary written from templates over recorded facts.
- **A site security score and daily intelligence** — where and in which hours situations begin, repeated vehicles and watchlist entries, and the share that turned out to be false alarms.
- **Feedback.** A person's review of what a closed situation turned out to be is kept, with the suggestion and the decision, as a dataset.
- **Off until asked.** A switch per tenant, off by default. If the background runner is stopped, alerts, incidents and video carry on exactly as before.
- *By design:* the background process cannot act — it reads, records and suggests, and tests hold that line. There is no language model. Nothing learns by itself: a rule or a weight changes only as a tenant setting or released code. A face that matched nobody is *unknown*, never *unauthorised*.

17. Smart Investigation

Finding what the platform already knows about a time, a place, a person or a vehicle — and keeping what was found.

- **One search across fourteen sources** — by period, site, camera, kind, event type, severity, risk, plate, name, member of staff and words. Each source is read under its own existing permission and the asker's sites; a source left out is named, with the reason.
- **A typed phrase, read by fixed rules** into that same search. The screen shows which words became which filter and which were not used, so the person can correct it. A phrase with nothing understood is refused rather than guessed at.
- **Investigations.** A file of why somebody was looking and what they found: references to records of any of the fourteen kinds, with notes, in the order they happened, each read live as the reader may see it.
- **The trail of a plate or a watchlist face** — every sighting, in order, with the time and the distance between.
- **Every search is on the record** with what was asked, not what was found.

- *Documented limitation:* an unidentified person is not followed across cameras, and the platform does not claim to. There is no search by a picture of a face.

ADDED SINCE EDITION 1 · MIGRATION 0144

18. Evidence packages and chain of custody

What was kept about a matter: put together, sealed, held against deletion, and accounted for.

- **What belongs is found and offered.** For an incident or an investigation: the frame of that detection, the recording of that camera at that moment, the drone media of that sighting. A person chooses what goes in.
- **Sealing.** A manifest of every item with its checksum is written, and the SHA-256 of the manifest. From then on a database trigger refuses every change.
- **Export** as a ZIP of the manifest as sealed and each original, byte for byte, with its checksum computed again and compared — plus a marked viewing copy of each picture.
- **One chain of custody per package,** from capture to release: captured, collected, accessed, sealed, placed under a hold, exported, downloaded, shared, released, hold lifted — each with who, in what role, and why.
- **Holds.** A hold stops all three retention jobs — frames and clips, recordings, drone media. Sealing places one on every item; a hold is lifted only by a person, with a reason.
- *Documented limitations:* video is not watermarked; nothing is digitally signed; the platform sends evidence to nobody — sharing and release are records of what a person did.

ADDED SINCE EDITION 1 · MIGRATION 0145

19. The security map

One operational map, and the places of a site as somebody who knows it drew them.

- **Ten layers on one map** — sites, cameras, guards, open incidents, live alerts, open situations, drones, patrol checkpoints, places and drone zones — each shown under its own screen's existing permission and the reader's sites. What has no position is counted under the map, not hidden.
- **The places of a site.** Buildings, floors, gates, access points, emergency and assembly points, zones and parking — a point, an outline or a level of a building, drawn by an administrator. Optional: a site without them works as before. A place is retired, never removed.
- **A door on the map.** An access point can name a door the platform knows, which gives door events a place to appear.
- **What is near.** Selecting an incident, alert or situation lists the cameras, drones, checkpoints and places within a radius, and every guard on shift at the site, nearest first.
- **Where a guard last was.** The last position each guard recorded this shift — a check-in, a scan, a status change — always with its age, and marked stale past an hour.
- *By decision:* guards are not tracked continuously. *Documented limitations:* there are no floor plans; distances are straight lines, not routes; alarm panels and sensors are not drawn.

20. Guard response, SLA and escalation

A sending as its own record: what the guard did about it, how long it took, and who is told when it is late.

- **The steps of a response.** Accepted, set off, arrived, reported — or cannot attend, with a reason — each written through to the incident's own status, history and arrival time. A guard who is not coming, or is stood down, gives the incident back.
- **The guard is told on their phone,** and answers there: Accept, On my way, I am there, Report what I found, I cannot attend.
- **Who to send — a suggestion with its reasons.** The guards on shift at the site, ranked by a score made of stated parts: free or already sent, distance by last recorded position and how old that position is, what they were already sent on this shift, and the certifications the site requires. A person chooses and dispatches.
- **Three clocks per incident** — acknowledge, arrive, resolve — judged every minute against the organisation's own settings. A clock that runs out is recorded once, marks the incident as breached, and is told to the person the settings name.
- **Escalation policies.** When an incident is still not acknowledged, reached or resolved after so long, tell a role or a person — by site and severity. Each step is recorded once, with how many people it reached.
- **A response desk** showing what each open incident is waiting for, with the procedure for it beside it.
- *By design:* the clocks are off until an organisation switches them on, and then apply only to incidents opened since. Nothing dispatches, reassigns or re-dispatches by itself.

21. The occurrence book: reviewed, corrected and handed on

The book read back — and what one shift hands the next.

- **Search** by words, kind, site, author, shift, period, where its review stands, and whether it has been corrected.
- **Supervisor review.** An entry is noted, marked to be followed up, or its follow-up closed with what was done — by somebody other than who wrote it. Every review is kept. What a reviewer wrote is shown to the people who keep the book, not to a client or a viewer.
- **Correction by a further entry.** A mistake is put right by a new entry that records which entry it corrects and why. The first stays exactly as written.
- **Two more kinds of entry** — a delivery, and unusual activity.
- **Instructions in force at a site.** Issued with or without a date they run out on, read by each guard, closed with a reason. The guards on shift are told, and the instruction is carried into every shift's summary until it ends.
- **A shift's summary, drafted by the platform and confirmed by a person.** Written from what was recorded during the shift — the book, incidents, alerts, patrols, dispatches, visitors, what the site has in hand, instructions, follow-ups — in fixed sentences that count and quote. The guard reads it, corrects it and confirms it; what the platform drafted is kept beside it, and once confirmed it cannot be changed.
- *By design:* the summary writes no prose of its own and draws no conclusion; no language model is involved.

22. The SOP library

Procedures in versions, approved before they are in force, and found by their own words.

- **A procedure and its versions.** A code it is cited by, a category, a site or every site. Versions are kept: drafted, submitted, approved to be in force from a date, or rejected with a reason.
- **Approved by somebody other than its author.** An approved version cannot be changed afterwards — a database trigger holds it.
- **In force, and running out.** Of the approved versions whose date has come, the latest is in force. A version that has run out leaves nothing in force rather than falling back to an older one.
- **The document as issued** can be attached to a version, kept with its SHA-256.
- **Asking the library** returns the passages of procedures in force that use the words asked — word for word, each with its procedure, its version and who approved it. When no passage uses those words it says so.
- **The procedure beside the incident.** A procedure names the kinds of incident it is for; the ones in force for an incident of that kind at its site are put beside it — on the response desk, on the phone's incident screen, and on the situation screen.
- *By design:* nothing composes an answer and no language model is involved — the platform never invents a procedure. *Documented limitations:* an attached PDF is kept and not read; the search reads English.

23. Visitor and contractor authorisation

Who said a visit may happen, for where, for how long and with whom — read at the gate, decided by the host.

- **An authorisation of a visit or a work permit.** Asked for by whoever registers visitors; approved or declined by the host, or by somebody who manages visits when no host is named. A no says why.
- **For a period, and for places.** Valid from and until; for named places of the site map or for the site in general. Where it stands — waiting, valid, run out, never answered — is its answer and the hour, worked out when it is read.
- **Escort and ID.** Whether the visitor is to be escorted, and by whom; and the kind of ID a named person says they saw. The number on an ID is not taken, and the one on the visit is never returned.
- **What the gate reads.** What stands on the record, as sentences made from it. It informs: checking a visitor in is unchanged, and does not look at the authorisation.
- **Cancelled or extended** by the host or somebody who manages visits, each with a reason. The host is told on their phone when asked; whoever asked is told the answer.
- **Where a badge was used.** The door events of the badge a visitor was given, while they had it, are set against the places and the period the visit is authorised for. One outside either is listed for somebody who manages visits to look at, who records that it was in order or how it was followed up.
- **On the phone:** the visits waiting for your answer, and what stands for a visit beside its check-in.
- *By design:* an authorisation admits nobody and refuses nobody, and a door event outside it is not a finding — nothing is raised and nobody is accused. *Documented limitations:* a visitor is followed only through doors their badge number opened; the people working under a work permit are not followed; these events are not fed to the intelligence layer; nothing opens or locks a door.

24. Device health, assets and maintenance

The health of devices as they report it, a register of what the organisation owns, and the work that keeps it working.

- **One reading of every device** — camera, recorder, sensor, drone, edge gateway, alarm panel — from what each reports: working, degraded, down, not known or switched off, with why and since when.
- **What is not known is not called working.** A camera with no stream, a recorder not probed for a day, a device that has never reported. A sensor reading a dangerous value is a working sensor.
- **What is kept.** Every five minutes each device's state is read and kept when it has changed: the states it has been in, and how long it was down in the last week and month.
- **The asset register.** A code, a kind, make, model, serial number, where it is, vendor, installed, warranty, status. An asset may be the device it is and then shows that device's health; a UPS, a server or a switch is an asset with no reading, and is shown as that. Known devices are put into the register in one step. Retired, never removed.
- **Work orders.** Raised on an asset, for a facility defect or for a site; given to one of the organisation's people or a named vendor; started; completed with what was done, the parts used and the time out of use; or cancelled with a reason.
- **Preventive schedules.** Something done every so many days puts its work forward before each date.
- **Suggested, then accepted.** A schedule falling due — and, once an organisation asks for it, a device read as down for a set number of hours — puts a work order forward. It is a suggestion until somebody who manages maintenance accepts it or dismisses it with a reason; it assigns nobody and tells nobody.
- *By design:* every answer and screen that shows a reading names what is not measured. *Documented limitations:* frame rate, latency, packet loss, the quality of the picture and gaps in a recording are not measured; an outage shorter than five minutes can pass unrecorded; no cause is inferred; there is no parts store, cost or maintenance SLA; the phone has no part of it.

25. Risk patterns and advice

Where what went wrong gathers — by hour, day, week and place — what stands out, how much history that rests on, and what a person made of it.

- **Five kinds of thing that went wrong, counted the same way:** incidents; doors refused, forced or tampered with; patrols missed or failed — a guard's tour, a virtual patrol, a drone patrol; devices going down; response clocks missed.
- **Counted three ways,** over the last one to twelve weeks: by weekday and hour where the site is, by week, and by place — a camera, a door, a patrol, a device. Nobody is named.
- **A week of hours** for each kind, shaded in four steps with a key that says in counts what each shade stands for; every cell readable without its shade, and the numbers themselves on request.
- **What stands out, by fixed rules:** a band of four hours, a weekday or a place holding most of the records; twice as many in the later half of the period as in the earlier; a device down three times or more. Each statement has its count in it and one thing to consider.
- **How much history it rests on.** Much, some or little: how many records, over how many weeks, and in how many of those weeks the same held by itself. One busy afternoon is one week, not four.
- **Nothing before is not called a rise.** When the earlier half of a period has no records, the statement says that the counts cannot tell a change from the start of recording.
- **A person's answer.** For one site: accepted, or not accepted with a reason. The platform counts the advice again first and keeps its own statement with the answer. An answer is added and never rewritten.

- *By design*: a pattern that recurred is not a forecast, and every reading says so; confidence is how much history a statement rests on, not a probability; an answer raises no work, moves no guard and alters no roster. *Documented limitations*: no risk score is made; nothing allows for how much is watched at a place, so more records there is not more danger there; a record made while testing is counted like any other; places are not zones; there is no report or export; the phone has no part of it.

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26. Platform operations — the vendor's console

Seventh AI sells this to security companies, so the vendor and the customer are different businesses and do not share a login, a permission set or a bill.

- **A platform tenant of its own.** The platform owner works from the vendor's tenant, not from inside a customer's, and holds none of a customer's day-to-day permissions — rosters, payroll, incidents, evidence. A customer's command centre is the customer's.
- **The customer's lifecycle** — trial, pending, active, suspended, expired, cancelled — rather than on or off, with a grace period after a subscription ends.
- **Pricing and licensing.** A catalogue of what each module costs and how it is charged, tied to the module licences the AI workers enforce.
- **The vendor's own invoices**, for customers who pay by bank transfer rather than by card.
- **Subscription notices** — a trial ending, a payment overdue, a licence about to lapse — each raised once, not nightly.
- **An error centre.** The same failure seen four thousand times is one row with a count: how many distinct things are broken, which customers each touches, since when, and whether anyone has dealt with it.
- **Support sessions, read-only by default.** The operator can look into a customer's account to answer a question without being able to change anything; elevated access is a separate, recorded choice.
- **Platform health and analytics** — recording, the intelligence runner, drone patrol; growth, adoption and churn from nightly snapshots.
- **The owner's account needs more than a password.** Two-factor authentication is enforced for the account that can reach every customer.
- *By design*: suspending a customer for non-payment is off by default. Cutting off a security company's cameras is a decision somebody makes deliberately, with their name on it.

Where AI is used, and where a person decides

"AI" covers several different things in this platform, and they are not trusted equally. This table says, for each place a model or a rule does work, what does it and what stays a person's. **It is the part of this document that is updated whenever an AI capability is added or upgraded** — with what the new capability does, what it is not trusted to do, and what a person still decides.

Where	What does the work today	What a person does
Detection on camera and drone video (11 modules)	Computer-vision models on frames: object detectors, a face embedding model, plate reading	Reads the alert; acknowledges, escalates or dismisses it
Face and plate watchlists	Embedding similarity in the database (pgvector); plate text matching	Keeps the watchlists; judges a match
Roster auto-scheduler	Scores candidates against nine stated rules. Not a learned model	Reviews the draft, edits it, publishes it
Security intelligence (module 16)	Rules, versioned as a named engine: relate events into situations, score how unusual and how serious, suggest a response with its reasons	Decides. Only a person's decision causes an action
Drone events (module 15)	Context rules and a risk engine over what the drone's camera detected; correlation with fixed cameras	Decides what becomes an incident; asks for a drone look
Who to send (module 20)	A score made of stated parts, shown with the suggestion	Chooses the guard and dispatches
A typed investigation phrase (module 17)	Fixed-rule parsing into a structured search, showing which words became which filter	Corrects the search; reads the results
Shift summary (module 21) and situation summary (module 16)	Templates: fixed sentences over recorded counts and quotations	Reads, corrects and confirms it
Finding a procedure (module 22)	Word search over approved text; passages quoted with their source	Reads the procedure and acts on it
A visitor's badge against what was authorised (module 23)	A comparison: the door events of the badge with the places and the period of the authorisation. Not a model	Looks at each one listed; records that it was in order or how it was followed up
Device health (module 24)	Rules over what each device reports: a stream's state, a probe, a heartbeat, a late reading. Nothing is inferred about the cause	Reads it; decides what is work
Work orders put forward (module 24)	Two rules: a schedule falling due, and — when an organisation has switched it on — a device read as down for a set number of hours	Accepts the suggestion as work, or dismisses it with a reason
Risk patterns and advice (module 25)	Fixed rules over counts of what was recorded: a band of hours, a weekday or a place holding most of the records; twice as many as before. Each says how much history it rests on. Not a model, and not a forecast	Reads it; answers it for a site — accepted, or not accepted with a reason. The answer changes nothing else
Daily intelligence and the site security score	Counting over a stated period of history	Reads it as history, not as a forecast

What the platform does not do today

- **No language model.** There is none in the platform and no dependency on one. Nothing is sent to an outside AI service. A typed question is parsed, not understood; a summary is a template, not prose.
- **Nothing decides for a person.** No part of the platform dispatches a guard, opens or closes an incident, flies a drone to an unplanned place, operates a door, admits or refuses a visitor, raises maintenance work, or changes a roster on a recommendation alone.
- **Nothing learns by itself.** A person's feedback on a situation is kept as a dataset; no model or rule is trained on it automatically. A rule or a weight changes only as a tenant setting or as released code.
- **Nobody is accused by a rule.** A face that matched nobody is *unknown*. An unidentified person is not followed across cameras. A prediction is a count over history, shown with the count under it.

In progress and planned

The enterprise expansion is fourteen phases, built and merged one at a time. Phases 1 to 9 are modules 17 to 25 above. What follows is not yet released and is **not counted** in any figure in this document. Each row moves into the numbered modules when it is merged and tested.

Phase	Area	State	What it adds
10	Operations analytics and the daily briefing	In progress	Boards per role; a dated daily briefing a person reviews and publishes; the reports that are missing
11	Workforce intelligence	Planned	One reading per guard and per site; training and coverage recommendations for a manager. Never an employment decision, never a change to a roster
12	Case management	Planned	Cases from incidents and investigations: investigators, tasks, notes, linked evidence, approvals, a report
13	Compliance and hardening	Planned	One statement of every retention period in force; a report of what is held about a subject; a security sweep over everything the expansion added
14	Integration testing	Planned	The eleven end-to-end chains — from CCTV to incident, incident to evidence, evidence to case — as tests

Waiting on something outside the code

- **A phone build and a device pass.** The phone parts of guard response, handover notes, procedures and visitor authorisation are type-checked and their rules tested, but have not been run on a device since the app moved to Expo SDK 57. Guards have them after a new build.
- **A new Windows desktop build.** Desktop 1.0.4 carries the screens up to AI Security Intelligence; the screens of modules 17 to 25 need a new build. The installer is not yet code-signed.
- **Real hardware.** A physical drone, access-control controllers and alarm panels have not been connected.
- **A cluster install.** The Helm chart lints and renders in CI and has not yet been installed on a cluster.

Client applications

Application	Platform	Scale	Notes
Web console	Browser (React)	106 routes	The full platform. Every module, every administrative surface.
Mobile app	iOS and Android (React Native, Expo SDK 57)	52 screens	Built for the guard on shift: check-in, patrol scanning, occurrence book, handover, post orders, incidents and the response to them, situations, drone events, SOS. Offline outbox queues actions taken with no signal and sends them when it returns.
Windows desktop	Electron	Same as web	Packaged installer for control-room machines, with auto-update. Version 1.0.4.

Roles

Eight built-in roles ship with the platform; tenants may define their own in addition. A role cannot grant a permission it does not itself hold. The permissions of modules 16 to 24 are held by the roles that work a site — not by Super Admin, and not by the Client role.

Role	Typical holder	Scope
Super Admin	Platform operator (the vendor)	The platform: customers, pricing, licences, errors, support sessions. Not a customer's operations
Admin	Agency owner or operations director	Everything within one tenant
Manager	Operations manager	Multi-site oversight and approvals
Supervisor	Site or shift supervisor	Rostering, attendance, review and incident response
Operator	Control room operator	Live monitoring, alerts, decisions and dispatch
Security Guard	Officer on shift	Mobile app: own shifts, patrols, responses and reports
Viewer	Auditor or observer	Read-only
Client	The agency's customer	Their own sites, incidents and invoices

Built for Singapore

- **Digital Occurrence Book** built against the Private Security Industry Act requirement for a licensed agency to maintain one. No screen or endpoint edits or removes an entry; a mistake is corrected by a further entry.
- **CPF** calculated by age band against the S\$7,400 Ordinary Wage ceiling, with eligibility determined by work pass type.
- **IR8A** annual reporting.
- **PWM grades** across the seven-grade Progressive Wage Model ladder, with wage floors that pay is checked against, and **PLRD licence** number and expiry per officer.
- **What a site requires an officer to hold**, with a daily report of who does not meet it.
- **Gazetted public holidays** with holiday pay, and the 72-hour monthly overtime cap flagged on every payroll run.

- **SGD invoicing** with purchase-order and payment-terms fields.
- **Work pass tracking** with expiry warnings.
- **PDPA** consent and data subject request handling; identity numbers kept to the minimum each record needs.
- **Asia/Singapore** as the tenant timezone, applied to every rendered date.
- **Four languages** reflecting the local workforce: English, Chinese, Malay and Tamil.

Known limits

Stated here in one place, as well as beside each module, because a reference that leaves them out is a brochure.

- **Drone Patrol has flown only in a simulator.**
- **Access control and alarm panels** have no real hardware behind them on the development system.
- **There is no language model.** Questions are parsed by fixed rules; summaries are templates; a procedure is found by its words.
- **An unidentified person cannot be followed across cameras.** A plate can; a watchlist entry can.
- **Guards are not tracked continuously.** A guard's position is the last one they recorded, shown with its age.
- **Device health is what devices report.** Stream state, disconnections, probes, heartbeats and late readings are read; frame rate, latency, packet loss, the quality of the picture and gaps in a recording are not measured, and an outage shorter than five minutes can pass unrecorded.
- **A visitor is followed only through the doors their badge opened,** and those door events are not fed to the intelligence layer.
- **A work order put forward by the platform** says what was read, not what is wrong: no cause is inferred for a device that is down.
- **Risk patterns are counts over history,** not forecasts. No risk score is made, nothing allows for how much is watched at a place, and a record made while testing is counted like any other.
- **Payroll:** graduated first- and second-year PR CPF rates and the Additional Wage ceiling are not implemented; PWM rates are not shipped.
- **Evidence:** video is not watermarked and nothing is digitally signed.
- **The phone app** needs a new build and a device pass for what was added after its SDK upgrade; **the desktop app** needs a new build for modules 17 to 25 and is not code-signed.
- **The Helm chart** has not been installed on a cluster.

Compiled from the Seventh AI Vision source repository at migration 0151 — 151 database migrations, 274 tables, 102 API modules serving 997 operations, 104 backend services, 106 web routes and 52 phone screens, covered by 5,920 automated tests (backend 3,955, repository inspection 1,276, web 406, phone 283). Capability counts reflect what is merged and running, not what is in progress or planned.